CHAPTER 1:

Introduction to the Cosmos

Some say the world will end in fire,
Some say in ice.
From what I've tasted of desire
I hold with those who favor fire.
But if it had to perish twice,
I think I know enough of hate
To say that for destruction ice
Is also great
And would suffice.

Robert Frost, 1920

The question of how the world will end has been the subject of speculation and debate among poets and philosophers throughout history. Of course, now, thanks to science, we know the answer: it's fire. Definitely, fire. In about five billion years, the Sun will swell to its red giant phase, engulf the orbit of Mercury and perhaps Venus, and leave the Earth a charred, lifeless, magma-covered rock. Even this sterile smoldering remnant is likely fated to eventually spiral into the Sun's outer layers and disperse its atoms in the churning atmosphere of the dying star.

So: fire. That's settled. Frost was right the first time.

But he wasn't thinking big enough. I'm a cosmologist. I study the universe, as a whole, on the largest scales. From that perspective, the world is a small sentimental speck of dust lost in a vast and varied universe. What matters to me, professionally and personally, is a bigger question: how will the *universe* end?

We know it had a beginning. About 13.8 billion years ago, the universe went from a state of unimaginable density, to an all-encompassing cosmic fireball, to a cooling, humming fluid of matter and energy, which laid down the seeds for the stars and galaxies we see around us today. Planets formed, galaxies collided, light filled the cosmos. A rocky planet orbiting an ordinary star near the edge of a spiral galaxy developed life, computers,